

AD 502 – Machine Learning

Pre-Reading Material – Lecture 15

K-Nearest Neighbor

Why This Topic Matters

How can we classify a new data point when similar examples are already available?

K-Nearest Neighbour (KNN) follows a simple idea: **similar data points are likely to belong to the same class**. It classifies a new instance by looking at its nearest labeled examples. KNN has applications in areas such as image recognition, recommendation systems, and pattern recognition.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the basic idea of KNN.
- Describe how KNN classifies a new data point.
- Explain the role of **k** and distance measures.
- Understand why feature normalization may be required.
- Explain why KNN is called a **lazy learner**.

1. How KNN Works

KNN starts with a labeled training dataset. For a new, unlabeled data point, it:

1. Finds the **k nearest data points**.
2. Examines their classes.
3. Assigns the new point to the class receiving the **majority vote**.

2. Measuring Similarity

KNN needs a **distance measure** to determine which data points are closest. **Euclidean distance** is traditionally used, while other measures such as **Manhattan distance** can also be considered.



3. Choosing the Value of k

The choice of k affects KNN's performance.

- **Small k :** more sensitive to noise and may cause overfitting.
- **Large k :** less sensitive to noise but may overlook important local patterns and cause underfitting.

Thus, selecting an appropriate k is important for achieving good generalization.

4. Importance of Normalization

Since KNN depends on distance, features with very different scales can affect the result unevenly. **Min-Max normalization** can be used to bring feature values into a common range, typically between 0 and 1.

5. KNN as a Lazy Learner

KNN is known as a **lazy or instance-based learning algorithm** because it mainly stores the training data rather than building an explicit model during training.

Therefore:

- Training is relatively fast.
- Prediction can take more time because neighbors must be identified when making a prediction.

Think Before the Lecture

1. Why does KNN need a distance measure?
2. What could happen if k is **too small**?
3. What could happen if k is **too large**?
4. Why is normalization useful in KNN?
5. Why is KNN called a **lazy learner**?